Programming for Problem Solving Using PythonUnit-1

Session-6: Data Types- Scalar Types

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Session-5: RECAP

Variables:

- Variables store data values.
- No explicit declaration needed.
- Examples:

```
x = 5
name = "Alice"
is_student = True
```

Session-5: RECAP

Numbers:

- Types: `int`, `float`, complex
- Integers (`int`): Whole numbers

```
a = 10
b = -3

```

- Floating-Point (`float`): Decimal numbers

```
pi = 3.14
e = 2.71

```

Session-5: RECAP

- Complex (complex): Real and imaginary part

```
z = 2 + 3j

```

Operations:

- Addition: +
- Subtraction: -
- Multiplication: *
- Division: /
- Floor Division: //

Session-5: RECAP

- Modulus: %
- Exponentiation: **

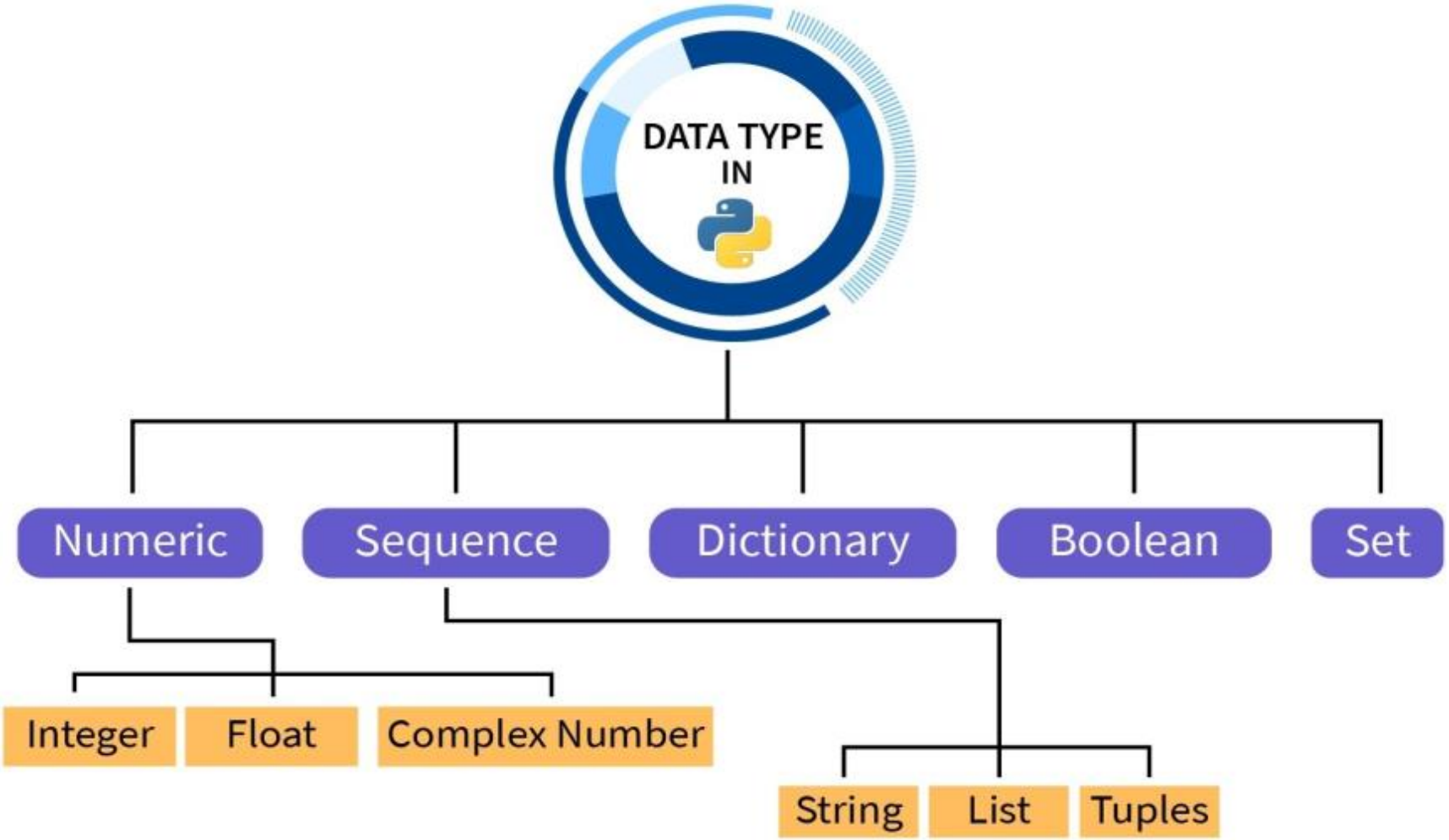
```
sum = 7 + 3      #10
difference = 10 - 4     #6
product = 2 * 3   #6
quotient = 15 / 5     #3.0
floor_div = 15 // 2     #7
remainder = 15 % 2     #1
power = 2 ** 3      #8

```

Session 6

Data Types- Scalar Types

Data Types: Scalar Types



Data Types: Scalar Types

In Python, scalar data types represent individual values. They are immutable, meaning once they are created, their values cannot be changed. Here are the main scalar data types in Python:

1. **Integer (int):** Represents whole numbers without decimals. Example: x = 5
2. **Float (float):** Represents numbers with decimals. Example: y = 3.14

Data Types: Scalar Types

3. **Complex (complex):** Represents numbers with a real and imaginary part. Example: z = 3 + 2j (where j represents the imaginary unit)

4. **Boolean (bool):** Represents truth values, either True or False. Example: is_valid = True

5. **String (str):** Represents an ordered sequence of characters. Example: message = "Hello, World!"

Data Types: Scalar Types

1. **Numeric Data types:** Numeric data types in Python represent numbers and are categorized into three types:
   - Integers in Python: Integers, as you know in mathematics are numbers without any fractional part. They can be 0, positive or negative. There is no limit to how long an integer can be in Python. They are represented by int class.

```
num = 5
print("number =", num)
```

Data Types: Scalar Types

- Floating Point numbers: Floating point numbers (float) are real numbers with floating point representation, they are specified by a decimal point. They are represented by the float class.

Syntax: Printing a floating point number in Python

```
num = 5.55
print("number =", num)
```

Data Types: Scalar Types

- Complex numbers: Complex numbers in python are specified as (real part) + (imaginary part)j. They are represented by the complex class. Complex numbers are of complex type. They are written in the form a+bj in which a is the real value and b is the imaginary value, where j represents the imaginary part.

Syntax: Printing a complex number in Python

```
num = 5 + 5j
print("number =", num)
```

Data Types: Scalar Types

In Python, the type() function is used to determine the data type of a value or a variable.

Program to demonstrate numerical values in Python

```
x = 10
y = 10.0
z = 10 + 10j
print("Data type of", x, ":", type(x))
print("Data type of", y, ":", type(y))
print("Data type of", z, ":", type(z))
```

Data Types: Scalar Types

OUTPUT:

Data type of 10 : $< class'int' >$
Data type of 10.0 : $< class'float' >$
Data type of (10+10j) : $< class'complex' >$

In the above example, we are using the type() function to show the type of data. It can be seen that ”10” is an int type, ”10.0” is a float type, and ”10+10j” is a complex type.

Data Types: Scalar Types

2. **Sequence Data types:** Sequence data types in Python are an ordered collection of similar or different values.
   – They are also called container data types as they usually contain more than one value.
   – We can access elements of a sequence data type by indexing. String, list, and tuple are the different types of containers used to store the data in a sequential manner.

Data Types: Scalar Types

- Strings in Python: A string can be defined as a sequence of characters enclosed in single, double, or triple quotation marks.
  – While in most cases single and double quotation marks are interchangeable.
  – Triple quotation marks are used for multi-line strings.
  – It is an immutable data type, i.e. its values cannot be updated.
  – A string is represented by a string class.

  Syntax: Strings are of string type and are represented by enclosing in quotation marks.

Data Types: Scalar Types

```python
# Python program to demonstrate strings.

# String with single quotes
x = 'Scaler'
print("x = ", x)
print("The data type of x:", type(x))

print()

# String with double quotes
y = "Scaler"
print("y =", y)
print("The data type of y:", type(y))

print()

# String with triple quotes (Multi-line strings)
z = '''Triple quotes are used
       for multi-line strings'''
print("z =", z)
print("The data type of z:", type(z))
```

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Data Types: Scalar Types

Output:

```
x = Scaler
The data type of x: <class 'str'>

y = Scaler
The data type of y: <class 'str'>

z = Triple quotes are used
        for multi-line strings
The data type of z: <class 'str'>
```

In the above example, we are demonstrating Python string, using single, double, and triple quotation marks. While single and double quotation marks are used for normal strings, triple quotation marks are used for multi-line strings.

Data Types: Scalar Types

Accessing elements of a string: Any character of a string can be accessed using indexing.

– Python also allows us to use negative indexing to access characters even from the back of a string.

– Note: Indexing of a sequence starts from 0.

– In the below example, we are accessing elements of the string using indexing.

– We are printing the first character of the string using the 0th index and the last character of the string using the -1th index (as Python supports negative indexing).

Scalar Types

```python
# Python program to access characters of a string.

my_string = "ScalerTopic"
print("String:", my_string)

# Printing first character of the string.
print("First character:", my_string[0])

# Printing last character of the string.
print("Last character:", my_string[-1])
```

Output:-

```
String: ScalerTopic
First character: S
Last character: c
```

Data Types: Scalar Types

- Lists in Python: Lists are an ordered sequence of one or more types of values.
  – They are just like arrays in other languages.
  – They are mutable, i.e. their items can be modified.

  Syntax: Lists are of the type list. They are created by enclosing items (separated by commas) inside square brackets [].

Data Types: Scalar Types

```python
# Python program to demonstrate lists.

my_list = [1, 10, "Scaler", 4, "A"]

print(my_list)
print("Its data type is", type(my_list))
```

Output:

```
[1, 10, 'Scaler', 4, 'A']
Its data type is <class 'list'>
```

Data Types: Scalar Types

Accessing elements of a list: Elements of a list can be accessed by referring to their index numbers, negative indexes to access the list items from its back.

– In the below example, we are accessing the elements of the list using indexing.

– At first, we are accessing the first element of the list by referring to its index "0" (Index number = Position of the element - 1) then we are referring to the fourth element by its index "3" and the last element of the list is accessed using negative indexing.

Data Types: Scalar Types

```python
# Python program to access elements of a list.

my_list = [1, 10, 'Scaler', 4, 'A']
print("List:", my_list)

# Accessing first element of the list.
print("First element of the list:", my_list[0])

# Accessing 4th element of the list.
print("4th element of the list:", my_list[3])

# Accessing last element of the list.
print("Last element of the list:", my_list[-1])
```

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Data Types: Scalar Types

Output:

```
List: [1, 10, 'Scaler', 4, 'A']
First element of the list: 1
4th element of the list: 4
Last element of the list: A
```

Data Types: Scalar Types

- Tuples in Python: Like lists, Tuples are also an ordered sequence of one or more types of values except that they are immutable, i.e their values can't be modified.
  – They are represented by the tuple class. *Syntax*:

  Tuples are of tuple type. They are created by a sequence of values separated by a comma with or without parentheses "()" for grouping of the sequence.

Data Types: Scalar Types

```python
# Python program to demonstrate tuples.

my_tuple = (1, 10, 'Scaler', 4, 'A')

print(my_tuple)
print("Its data type is", type(my_tuple))
```

Output:

```
(1, 10, 'Scaler', 4, 'A')
Its data type is <class 'tuple'>
```

Data Types: Scalar Types

In the above example, we are creating a tuple by enclosing the elements inside parentheses and then printing it. In the second line, we are printing its type using the type() function.

Data Types: Scalar Types

Accessing elements of a tuple: Elements of a tuple can be accessed by referring to their index numbers, negative indexes to access the tuple items from its back.

– In the above example, we are accessing the elements of a tuple using indexing.

– At first, we are printing the first element using the 0th index, then we are printing its third element using the 2nd index, at last, we are printing its last element using negative indexing.

Data Types: Scalar Types

```python
# Python program to access elements of a tuple.

my_tuple = (1, 10, 'Scaler', 4, 'A')
print("Tuple:", my_tuple)

# Accessing first element of the tuple.
print("First element of the tuple:", my_tuple[0])

# Accessing 3rd element of the tuple.
print("3rd element of the tuple:", my_tuple[2])

# Accessing last element of the tuple.
print("Last element of the tuple:", my_tuple[-1])
```

Output:

```
Tuple: (1, 10, 'Scaler', 4, 'A')
First element of the tuple: 1
3rd element of the tuple: Scaler
Last element of the tuple: A
```

Data Types: Scalar Types

3. **Dictionaries in Python:** In Python, Dictionaries are an unordered collection of key-value pairs (key: value).
   – This helps in retrieving the data and makes the retrieval highly optimized, especially in cases of high volume data.
   – Keys can't be repeated in a dictionary while values can be repeated.
   – Dictionaries are mutable, i.e. their values can be modified.

Data Types: Scalar Types

Syntax: Dictionaries are of the type dict.
– The elements of a dictionary are endosed in curly brackets , where each element is separated by a comma , and key-value pair by a colon :.
– In the below example, we are creating a dictionary by endosing the key-value pair in curly brackets then we are printing it.
– In the second line, we are printing its type using the type() function.

Data Types: Scalar Types

```python
# Python program for demonstrating dictionaries.

my_dict = {"Name": "Tom", "Age": 50, "Movie": "Mission Impossible"}

print(my_dict)
print("Its data type:", type(my_dict))
```

Output:

```
{'Name': 'Tom', 'Age': 50, 'Movie': 'Mission Impossible'}
Its data type: <class 'dict'>
```

Data Types: Scalar Types

Accessing values of a dictionary: Values of a dictionary are accessed by referring to their keys.

```
# Python program for accessing values of a dictionary.

my_dict = {'Name': 'Tom', 'Age': 50, 'Movie': 'Mission Impossible'}

print(my_dict["Name"])
print(my_dict["Age"])
print(my_dict["Movie"])
```

Data Types: Scalar Types

In the above example, we are accessing the values of a dictionary by referring to their keys, in the first line we are referring to Tom using the key Name, then we are referring to 50 using the key Age, at last, we are referring to Mission Impossible using the key Movie.

Output:

```
Tom
50
Mission Impossible
```

Data Types: Scalar Types

4. **Booleans in Python:** Boolean is a data type that has one of two possible values, True or False.
   – They are mostly used in creating the control flow of a program using conditional statements.

   Syntax: The True value in the boolean context is called "truthy" and False value is called "falsy".

   In the below example, we are printing the boolean variables a (True) and b (False), then we are printing their types using the type() function.

Data Types: Scalar Types

```python
# Python program to demonstrate boolean.

a = True
b = False

# Printing boolean values
print("a =", a)
print("b =", b)

print()

# Data type of True and False
print("Data type of 'True':", type(a))
print("Data type of 'False':", type(b))
```

Output:

```
a = True
b = False

Data type of 'True': <class 'bool'>
Data type of 'False': <class 'bool'>
```

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Data Types: Scalar Types

5. **Sets in Python:** Sets in Python are an unordered collection of elements.
   – They contain only unique elements. They are mutable, i.e. their values can be modified.

   Syntax: Sets are of the type set. They are created by elements separated by commas enclosed in curly brackets .

Data Types: Scalar Types

```python
# Python program for demonstrating sets.

my_set = {1, 8, "Scaler", "F", 0, 8}

print(my_set)
print("Its data type:", type(my_set))
```

Output:

```
{0, 1, 'F', 8, 'Scaler'}
Its data type: <class 'set'>
```

Data Types: Scalar Types

In the above example, we are creating a set by enclosing the elements inside curly brackets "" then we are printing it.

In the second line, we are printing its type using the type() function.

While creating the set, we have put two duplicate values 8 but when we printed it, 8 occurred only once, that's because sets only have unique elements.

Test Your Knowledge- Fill in the Blanks

1. In Python, the data type used to represent text is called a _____.
2. The data type used for whole numbers in Python is _____.
3. A data type in Python used for numbers with a decimal point is _____.
4. To store a sequence of characters in Python, we use a _____.
5. The data type used to represent true or false values in Python is _____.

Test Your Knowledge- Fill in the Blanks (Solutions)

1. In Python, the data type used to represent text is called a string.
2. The data type used for whole numbers in Python is int.
3. A data type in Python used for numbers with a decimal point is float.
4. To store a sequence of characters in Python, we use a string.
5. The data type used to represent true or false values in Python is boolean (bool).

Test Your Knowledge- MCQ

1. Which of the following is a scalar data type in Python?
   - a Integer
   - b List
   - c Dictionary
   - d Tuple
2. What does the 'float' data type represent in Python?
   - a Whole numbers without decimals
   - b Numbers with decimals
   - c Numbers with real and imaginary parts
   - d Boolean values

Test Your Knowledge- MCQ

3. Which data type is used to represent truth values in Python?

   a. Integer
   b. Float
   c. Complex
   d. Boolean

4. What is the correct syntax for defining a complex number in Python?

   a. `z = 3 + 2i`
   b. `z = 3 + 2j`
   c. `z = (3, 2)`
   d. `z = [3, 2]`

Test Your Knowledge- MCQ

1. Which of the following is a scalar data type in Python?
   - a. Integer
   - b. List
   - c. Dictionary
   - d. Tuple
2. What does the 'float' data type represent in Python?
   - a. Whole numbers without decimals
   - b. Numbers with decimals
   - c. Numbers with real and imaginary parts
   - d. Boolean values

Test Your Knowledge- MCQ

3. Which data type is used to represent truth values in Python?

   a. Integer
   b. Float
   c. Complex
   d. Boolean

4. What is the correct syntax for defining a complex number in Python?

   a. `z = 3 + 2i`
   b. `z = 3 + 2j`
   c. `z = (3, 2)`
   d. `z = [3, 2]`

Test Your Knowledge- Interview Questions

1. What are scalar data types in Python?
2. Explain the difference between integers and floats in Python.
3. How do you define a complex number in Python?
4. What is the purpose of the boolean data type in Python?
5. How do you concatenate two strings in Python?

Test Your Knowledge- Interview Questions (Solutions)

1. *Ans*: Scalar data types in Python represent individual values and are immutable. Examples include integers, floats, complex numbers, booleans, and strings.
2. *Ans*: Integers represent whole numbers without decimals, while floats represent numbers with decimals. For example, 5 is an integer, and 3.14 is a float.
3. *Ans*: Complex numbers in Python are defined using the syntax `z = a + bj`, where `a` is the real part, and `b` is the imaginary part. For example, `z = 3 + 2j`.

Test Your Knowledge- Interview Questions (Solutions)

4. *Ans:* The boolean data type in Python is used to represent truth values, which can be either `True` or `False`. It is commonly used for logical operations and conditional statements.
5. *Ans:* Strings in Python can be concatenated using the '+' operator. For example, `"Hello" + "World"` will result in `"HelloWorld"`.

Thank You!

Any Questions?